



Is Grokking a Loss of Normal Hyperbolicity of the Interpolation Manifold?

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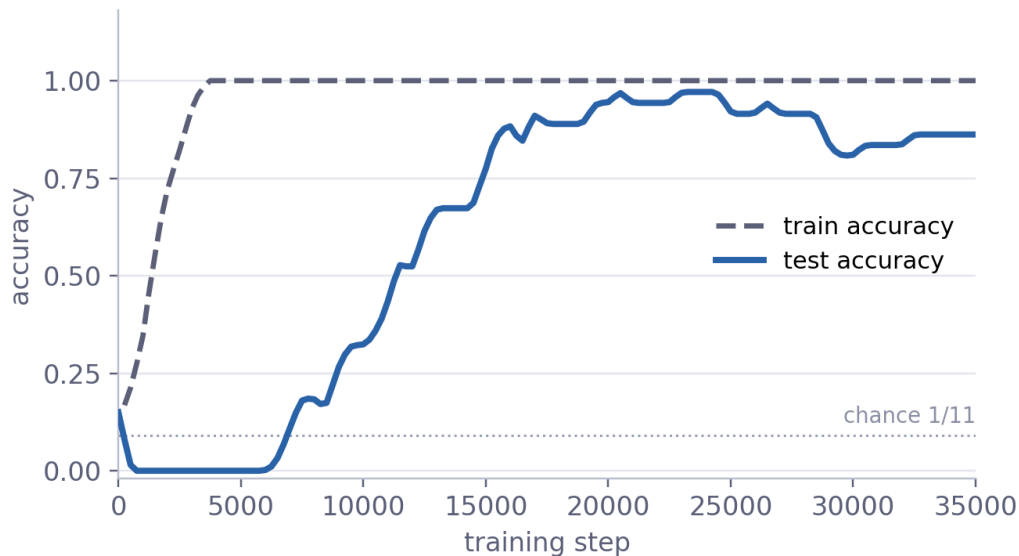
What is grokking? An example from our run

Overview

1 Memorization

2 Plateau

3 Transition



Grokking

A network first fits its training set perfectly while test accuracy stays at or below chance — and generalizes only much later [Power et al. 2022].

Click a phase to step through it.

Two-layer ReLU network learning $(a + b) \bmod 11$ from 70 % of all pairs; setup details later. Curves traced from Fig. 1 of the paper (seed 0).

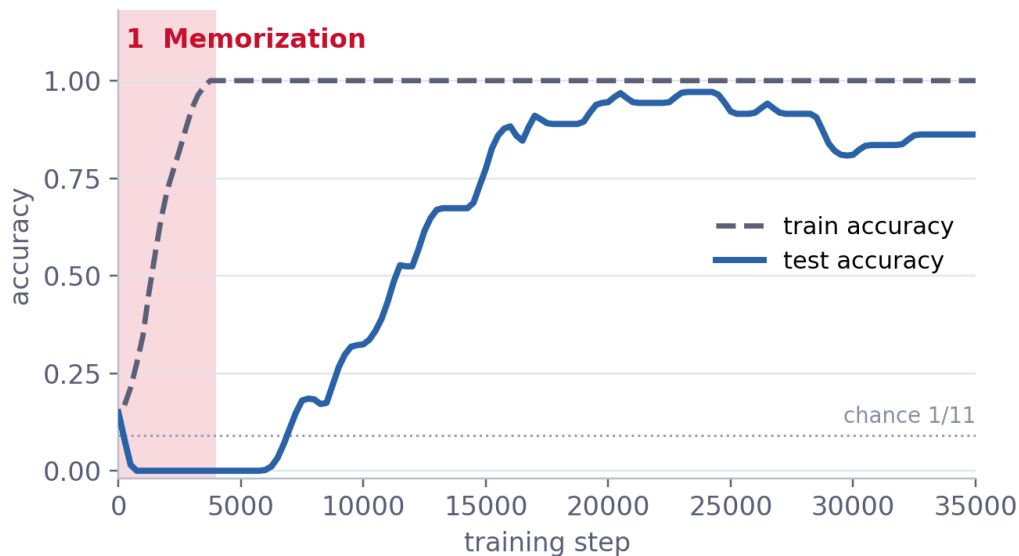
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1 Memorization

Training accuracy climbs from chance to 1.0 within about 4,000 steps: the network fits every training pair.

Test accuracy falls to 0.0 — it is learning the training set, not the rule.

Two-layer ReLU network learning $(a + b) \bmod 11$ from 70 % of all pairs; setup details later. Curves traced from Fig. 1 of the paper (seed 0).

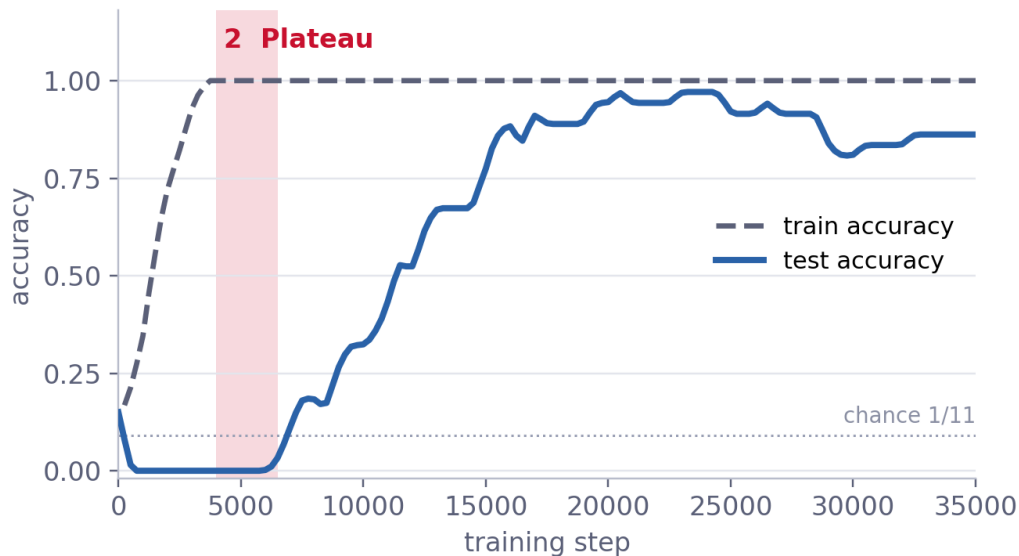
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2 Plateau

Training accuracy 1.0, test accuracy 0.0 — below the $1/11 \approx 0.09$ chance level.

The memorizing solution actively mispredicts unseen pairs. In the accuracy curves, nothing seems to happen for a while ($\approx$ steps 4,000–6,000).

Two-layer ReLU network learning $(a + b) \bmod 11$ from 70 % of all pairs; setup details later. Curves traced from Fig. 1 of the paper (seed 0).

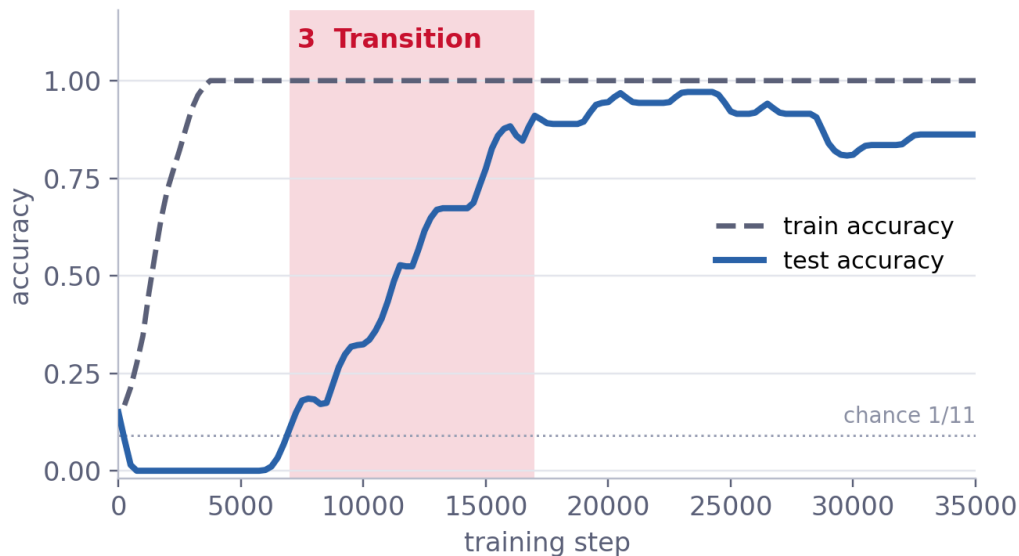
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3 Transition

Test accuracy rises over roughly steps 7,000–17,000 to ≈ 0.95 , then settles near 0.86 — with training accuracy unchanged.

This talk: what happens geometrically in this window?

Two-layer ReLU network learning $(a + b) \bmod 11$ from 70 % of all pairs; setup details later. Curves traced from Fig. 1 of the paper (seed 0).

Current account: interpolation first, then slow norm drift

$$d\theta/dt = -\nabla L(\theta) - \lambda\theta$$

Fast: $-\nabla L$ pulls the parameters onto the zero-loss (interpolation) manifold.

Slow: $-\lambda\theta$ (small weight decay) drifts along that manifold toward lower norm.

Gradient-flow limit with decoupled weight decay λ .

Muřat 2025

Norm minimization on the zero-loss manifold, in the limit of infinitesimal learning rate and weight decay.

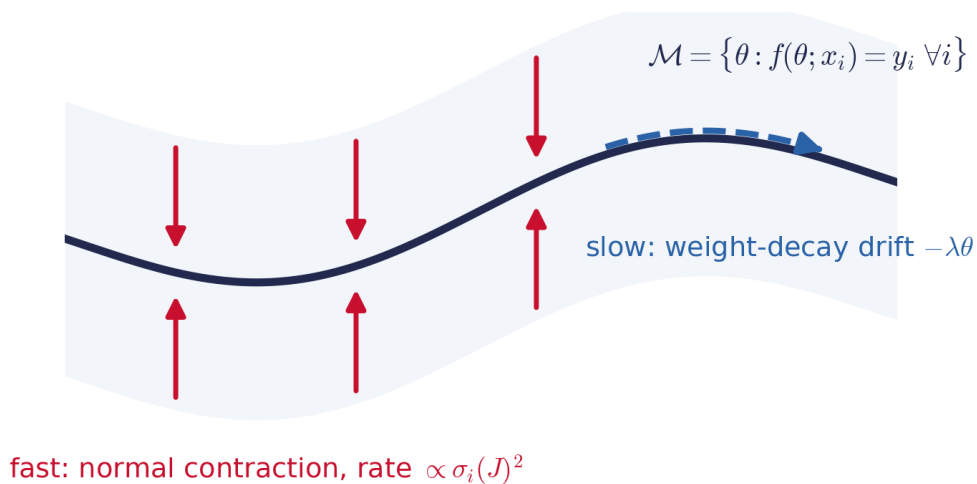
Boursier, Pesme & Dragomir 2025

Same endpoint via a Riemannian norm flow on the manifold of interpolating solutions.

Lyu et al. 2024

Homogeneous nets, large init, small weight decay: provable sharp transition from a kernel predictor to a min-norm / max-margin KKT point.

Seen as dynamics: a fast–slow system with a slow manifold



Schematic, not data.

The interpolation manifold $\mathcal{M}$ plays the role of a slow manifold.

Normal hyperbolicity: normal contraction rates stay bounded away from 0 along the slow drift.

Fenichel theory [Fenichel 1979; Kuehn 2015]: then the slow manifold persists and the reduced flow is well defined.

Gap: the grokking accounts share this structure but do not invoke this mathematics — nor ask what happens when normal hyperbolicity is lost.

Research question

Is the sharp generalization transition in grokking a **loss of normal hyperbolicity** of the interpolation manifold — or does the manifold stay **uniformly attracting** while generalization happens by smooth drift?

What we set out to do

1 Formalize

Cast post-memorization training as a fast–slow system with the interpolation manifold as slow manifold, and state two competing hypotheses.

2 Measure

Derive one scalar, optimizer-agnostic diagnostic that would detect a loss of normal hyperbolicity.

3 Test

Track it through a real grokking run against test accuracy — across 5 seeds and the low end of the spectrum.

Two hypotheses for the sharp transition

H1 — bifurcation

A normal restoring direction softens toward 0; the timescale separation breaks and the trajectory jumps (fold / pitchfork). Modular addition's symmetry makes a pitchfork a natural candidate.

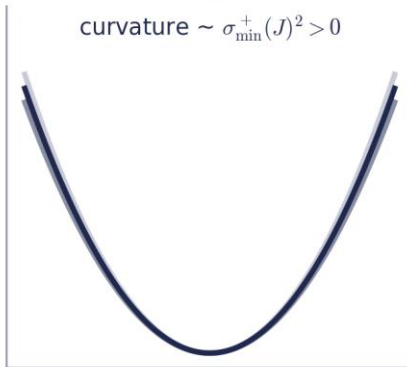
Prediction: $\sigma_{\min}^+(J)$ dips toward 0 at the transition

H0 — smooth contraction

The manifold stays uniformly normally hyperbolic; the transition is a feature of the slow drift through a region where test behaviour changes quickly.

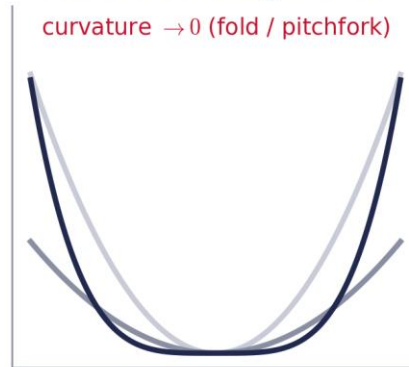
Prediction: $\sigma_{\min}^+(J)$ stays bounded away from 0

H0: rate stays bounded
curvature $\sim \sigma_{\min}^+(J)^2 > 0$



normal coordinate

H1: a direction goes flat
curvature $\rightarrow 0$ (fold / pitchfork)



normal coordinate

schematic, not data · light → dark = earlier → later in training

The diagnostic: a single scalar along the trajectory

Squared loss $L(\theta) = (1/n) \sum_i \|f(\theta; x_i) - y_i\|^2$

Residuals $r(\theta) = (f(\theta; x_i) - y_i)_i \in \mathbb{R}^{nc}$

Residual Jacobian $J(\theta) = \partial r / \partial \theta \in \mathbb{R}^{nc \times D}$, Gauss–Newton $G = J^T J$

On $\mathcal{M}$: $\text{row}(J)$ = normal space, $\ker J$ = tangent space

Why squared loss? It makes the interpolation manifold an exact object with a well-defined tangent and normal structure.

slowest normal restoring rate

$$\kappa(\theta) = \sigma_{\min}^+(J(\theta))^2$$

$\sigma_{\min}^+(J)$ = smallest nonzero singular value of J . With $nc < D$, J generically has full row rank, so all nc singular values are positive.

Why $\sigma_{\min}^+(J)$ measures normal contraction

1 At an interpolating point

$$\begin{aligned} r(\theta^\star) &= 0 \Rightarrow \\ \nabla^2 L &= (2/n)(J^\top J + \sum_k r_k \nabla^2 r_k) \\ &= (2/n) J^\top J \end{aligned}$$

2 Linearized fast flow

$d\delta\theta/dt \approx -(2/n) J^\top J \delta\theta$
Tangential parts ($\ker J$) are not restored; normal parts contract at rates $(2/n) \sigma_i(J)^2$

3 Normal hyperbolicity

$\Leftrightarrow$ the slowest rate stays bounded away from 0 along the drift.

Loss of normal hyperbolicity $\Leftrightarrow$
 $\sigma_{\min}^+(J) \rightarrow 0$

Read $\sigma_{\min}^+(J(\theta(t)))$ against test accuracy: **dip toward 0 $\rightarrow$ H1** **bounded away from 0 $\rightarrow$ H0**

Depends only on the model and data at $\vartheta \rightarrow$ well defined under any optimizer, not only gradient flow.

Experimental setup



Task

Modular addition mod $p = 11$; one-hot (a, b) in, one-hot $(a + b) \bmod p$ out; random 70 / 30 train/test split.



Model

Two-layer ReLU network, width 96, no second-layer bias.



Loss & init

Squared loss, full batch; large init (Kaiming $\times 3.5$) for a clear memorization plateau [Liu et al. 2023].



Optimizer

AdamW, $\text{lr } 3 \times 10^{-3}$, decoupled weight decay 2.0, 35,000 steps.



Measurement

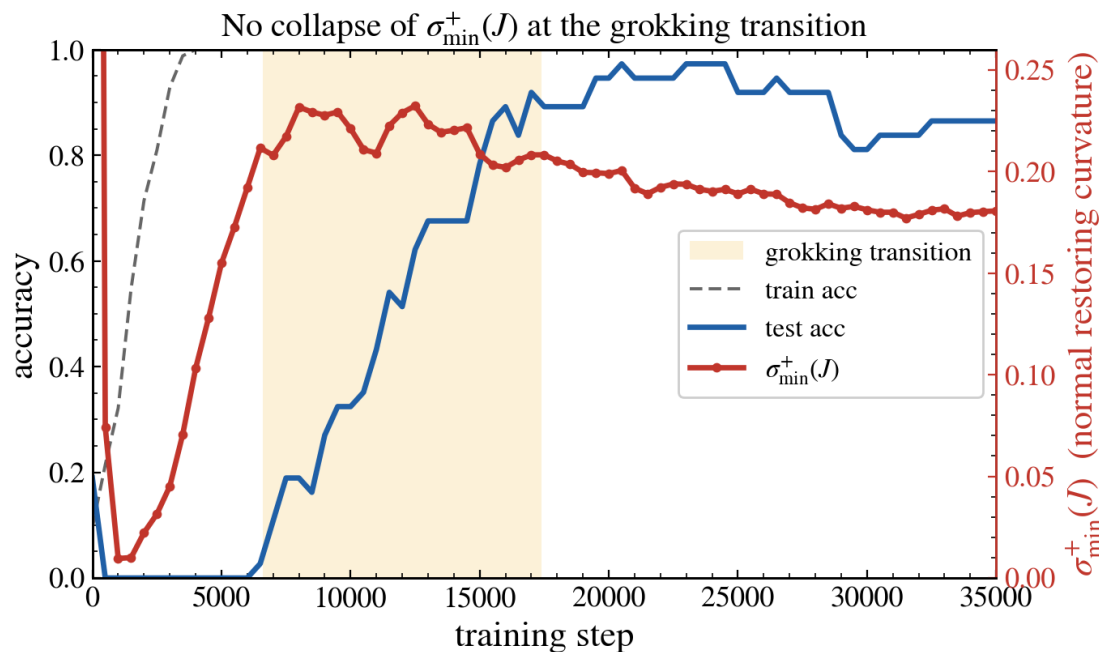
Snapshot θ every 500 steps; SVD of J in float32 in a separate pass.



Robustness

5 initialization seeds; the 6 smallest singular values tracked.

Result: $\sigma_{\min}^+(J)$ does not collapse at the transition



Same run as the grokking example, now with $\sigma_{\min}^+(J)$ in red.

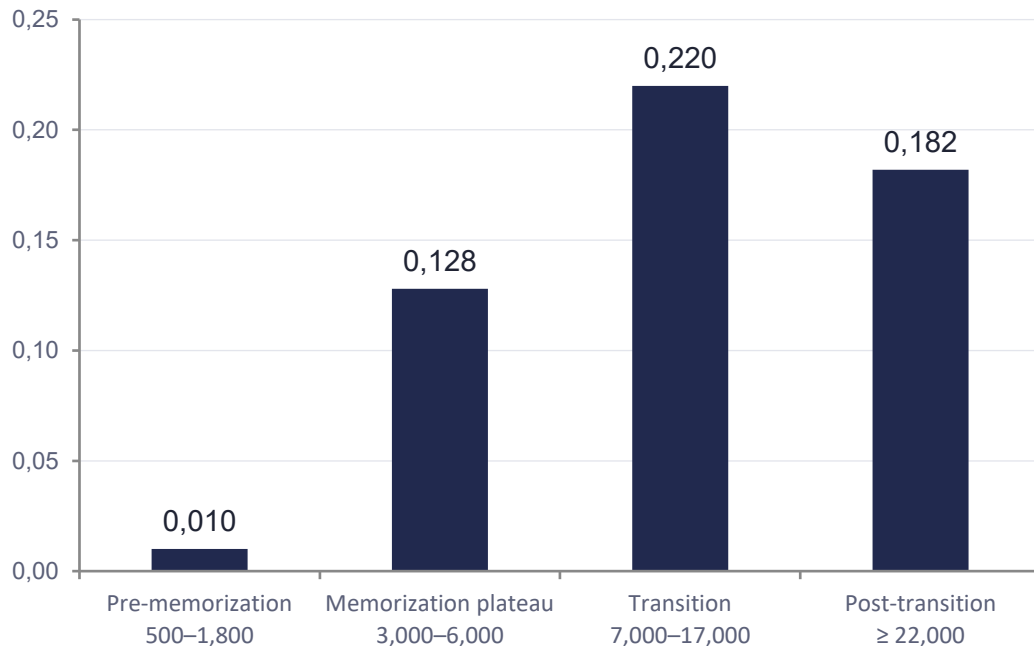
$\sigma_{\min}^+(J)$ near 0 only before memorization: global min $\approx 1 \times 10^{-2}$ at step 1,000.

During the transition it is at its largest values — the restoring rate peaks rather than collapsing.

Fig. 1 of the paper · single seed · shaded band = generalization transition

By phase: highest restoring rate during generalization

Median $\sigma^+ \min(J)$ by training phase (single seed)



Transition window

min 0.202, median 0.220 — the highest of all phases.

Only near-zero regime

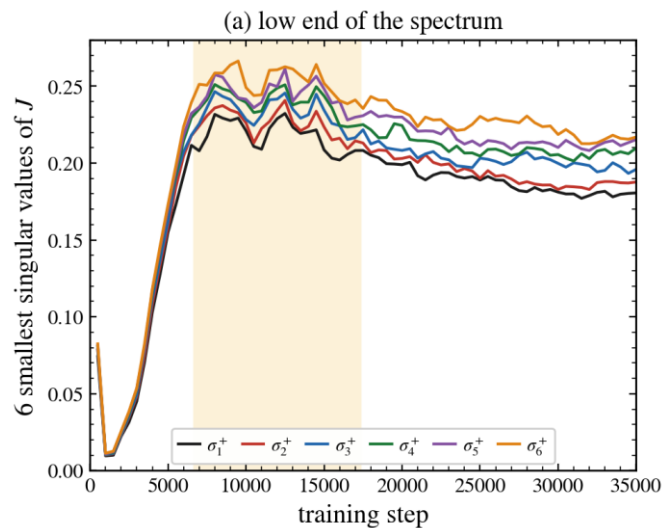
Pre-memorization (min 0.0096) — unrelated to grokking.

Instrument check

That reading shows the diagnostic can register a genuinely near-degenerate Jacobian when one occurs.

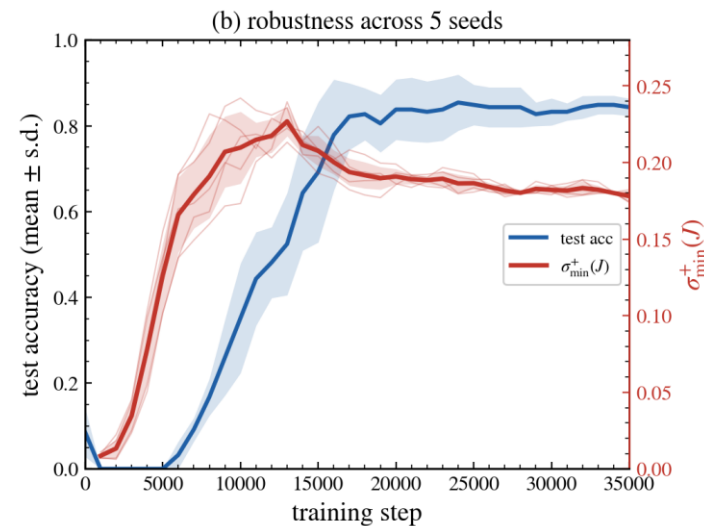
Two checks: not one singular value, not one seed

(a) Low end of the spectrum



The six smallest singular values form a tight cluster bounded away from 0 — no low-dimensional subspace collapses.

(b) Five seeds



Transitions at different steps (test acc > 0.5 between ≈ 10 k and 16k), yet in every seed $\sigma_{\min}^+(J)$ stays in 0.18–0.23 in its transition window.

Side observation: no visible ℓ_2 -norm contraction (AdamW)

≈ 10

parameter norm through the transition — it does not decrease, and drifts slightly upward

Interpretation

Consistent with AdamW's implicit bias being governed by an ℓ_∞ -type rather than ℓ_2 geometry [Xie & Li 2024; Gronich & Vardi 2026]. We treat it as a caveat about the optimizer, not a claim about the manifold.

What we deliberately do not report

The timescale ratio $\sigma_{\min}^+(J)^2 / \lambda$. In the gradient-flow theory λ is the drift rate; AdamW's decoupled weight decay does not play that role (Adam rescales the effective step per coordinate), so the raw ratio would misstate the separation.

What this does not establish



Optimizer

AdamW, while the fast-slow theory is stated for gradient descent / flow. The sharpest regimes (GD, large init, small weight decay) are untested.



Sharpness

The observed transition is moderately gradual rather than maximally sharp.



Global measure

A loss of normal hyperbolicity confined to a narrow, test-relevant subspace (e.g. symmetry-aligned directions) could hide from the low end of the spectrum.



Discrete trajectory

We measure snapshots of a discrete trajectory, not the gradient-flow limit in which the manifold is exactly defined.

Framing: constraining, not refuting — in this regime, grokking is not accompanied by a global collapse of the manifold's normal restoring rate.

Conclusion and next steps

Diagnostic: $\sigma_{\min}^+(\mathbf{J})$ turns the bifurcation question into a single measurable curve.

Finding: the manifold stays normally hyperbolic through the transition — across 5 seeds and the 6 smallest singular values.

Reading: evidence favours H0 (smooth contraction) over H1 (bifurcation) in this regime.

Next experiments

1

Same diagnostic under gradient descent in a sharp-transition regime.

2

Subspace-resolved curvature: project onto directions that best separate train from test, or onto symmetry-aligned subspaces.

3

If no dip appears: characterize which geometric quantity does move monotonically — under AdamW possibly an ℓ_∞ - or sparsity-based measure.

Thank you — questions?

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github.com/baksho/grokking-normal-hyperbolicity



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Backup Slides

Backup — click a question to jump to it

[Q1 Why AdamW, when the fast–slow theory is stated for gradient descent?](#)

[Q2 Grokking is usually studied with cross-entropy. Why squared loss?](#)

[Q3 Could a loss of normal hyperbolicity hide in a subspace you did not look at?](#)

[Q4 ReLU makes \$J\$ piecewise constant and you sample every 500 steps. Could you miss a short dip?](#)

[Q5 The derivation assumes \$r = 0\$. Is the network actually on the manifold?](#)

[Q6 If the manifold stays normally hyperbolic, what makes the transition sharp?](#)

[Q7 Does this hold beyond a tiny setting \(\$p = 11\$, two-layer MLP\)?](#)

Why AdamW, when the fast–slow theory is stated for gradient descent?

Short answer

Practicality: AdamW reached the transition within our compute budget. The diagnostic itself does not depend on the optimizer.

Optimizer-agnostic: $\sigma_{\min}^+(J)$ depends only on the model and data at θ , so it is well defined for any trajectory.

What does change: the interpretation of the slow drift. Under AdamW, λ is not the drift rate — which is why we do not report $\sigma_{\min}^+(J)^2/\lambda$.

Open: the sharpest regimes (GD, large init, small weight decay) are untested — our first next experiment.

Grokking is usually studied with cross-entropy. Why squared loss?

Short answer

Squared loss makes the interpolation manifold an exact object: $M = \{\theta : f(\theta; x_i) = y_i \text{ for all } i\}$, with a well-defined tangent and normal structure.

With squared loss, $r = 0$ on M , so the Hessian there is exactly the Gauss–Newton matrix — the whole derivation rests on this.

With cross-entropy on separable data the loss has no finite minimizer (shown for linear models by Soudry et al. 2018): it only approaches 0 as the logits grow, so there is no exact zero-loss set to be normally hyperbolic.

Lyu et al. 2024 obtain their grokking transition for both classification and regression, so squared loss is not an exotic setting.

Could a loss of normal hyperbolicity hide in a subspace you did not look at?

Short answer

Yes, in principle. This is the main limitation we state, and the most important follow-up.

We already go beyond one number: the **six** smallest singular values stay clustered and bounded away from 0.

But they are still a global measure: a softening confined to a narrow, test-relevant subspace (e.g. symmetry-aligned directions of modular addition) could leave the low end of the spectrum untouched.

Needed: a subspace-resolved curvature — J projected onto the directions that most separate train from test points, or onto symmetry-aligned subspaces.

ReLU makes J piecewise constant and you sample every 500 steps. Could you miss a short dip?

Short answer

A dip shorter than the snapshot spacing could in principle fall between snapshots; denser sampling around the transition is a cheap check.

J changes when activation patterns switch; in practice the $\sigma_{\min}^+(J)$ trajectory is smooth at snapshot resolution.

The transition spans $\approx 7,000$ – $17,000$ steps, i.e. about 20 snapshots, and shows no downward trend at all — the values are at their maximum.

My expectation (not tested): approaching a fold, the normal rate would soften progressively rather than in one isolated step — which would be visible as a trend.

The derivation assumes $r = 0$. Is the network actually on the manifold?

Short answer

Not exactly: with weight decay and a discrete optimizer the iterate sits near M , not on it. The identity $\nabla^2 L = (2/n) J^T J$ is exact only at $r = 0$.

The neglected term $(2/n) \sum_k r_k \nabla^2 r_k$ is weighted by the residuals, so it is small when the training residuals are small.

The paper does not report the residual magnitude.

This is part of the fourth limitation: a discrete trajectory, not the gradient-flow limit in which M is exactly defined.

If the manifold stays normally hyperbolic, what makes the transition sharp?

Short answer

H0: the geometry stays regular; the sharpness comes from the slow drift passing through a region where test behaviour changes quickly.

This matches the smooth-contraction picture [Boursier et al. 2025; Lyu et al. 2024; Muşat 2025].

Related: Truong et al. 2026 frame the grokking delay as norm contraction across a geometric gap (cited for context; not reproduced by us).

Caveat for our run: the ℓ_2 norm does not shrink under AdamW, so identifying the quantity that does move monotonically is open — possibly ℓ_∞ - or sparsity-based.

Does this hold beyond a tiny setting ($p = 11$, two-layer MLP)?

Short answer

We do not know — the claim is deliberately scoped to this setting.

Here $n = 84$ training pairs, $c = 11$ classes, so J is $924 \times 3,264$ ($nc < D$, full row rank generically) — a full SVD is cheap.

For larger models a full SVD becomes expensive; only the smallest singular values are needed, so iterative methods would be the natural route (not something we have implemented).

Natural next settings: larger p , deeper networks, transformers — together with GD and a sharper transition.